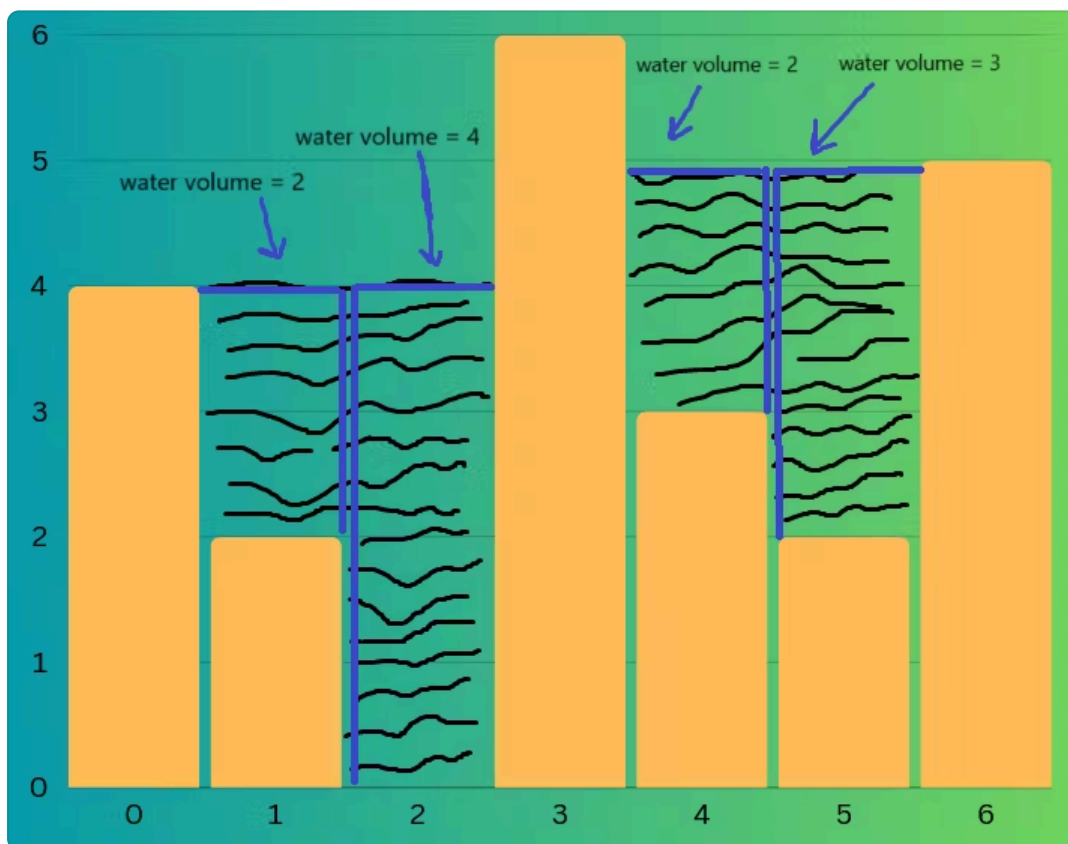




Trapping Rainwater problem by kanha_ji



The Trapping Rainwater problem is an algorithmic challenge where one must calculate the total amount of rainwater that could be trapped between bars of varying heights, which represent buildings. The array should have at least three elements, and the elements should not be in strict ascending or descending order to allow for water to be trapped. Two auxiliary arrays are constructed to store the highest bars to the left and right of each bar. For a given array, these auxiliary arrays are populated by iterating from left to right and right to left, respectively, determining the maximum height encountered so far. The trapped water at each position is then calculated by taking the minimum of the corresponding values in the auxiliary arrays, subtracting the height of the current bar, and adding it to the total volume. Negative water volumes are considered invalid and set to zero. The process is repeated for each bar to find the cumulative trapped water volume.



Trapping rainwater is a common question in data structure and algorithms.

Imagine a chart with bars of different heights. These bars are like buildings. When it rains, water can get trapped between two buildings (or bars).

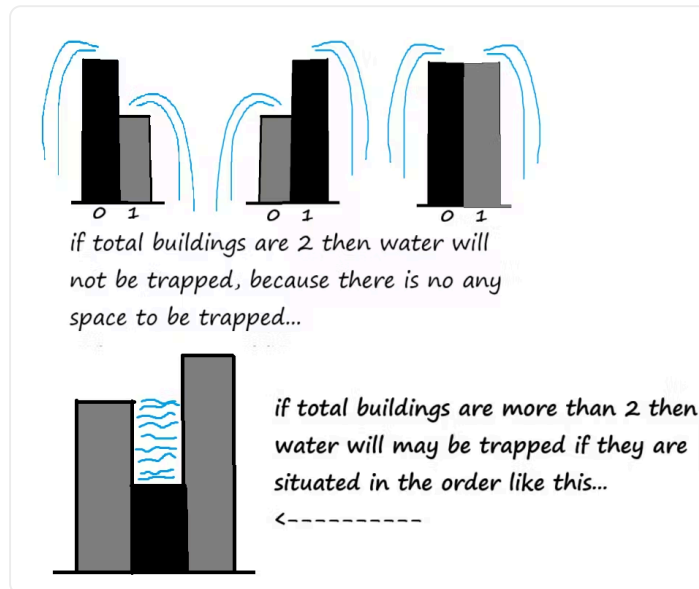
The goal is to find out how much water is trapped between the buildings.

let's understand it deeply using the chart above and multiple charts ahead...

first of all, we need to know some basic rules related to this problem.

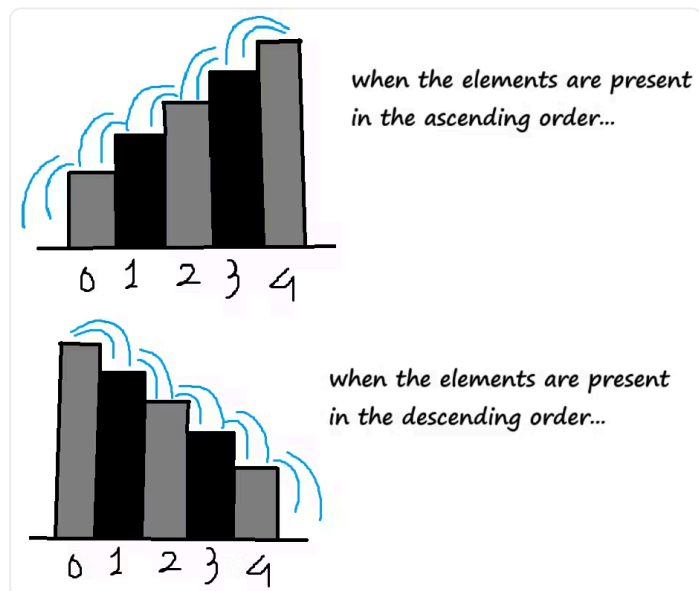
1) there should be more than 2 elements in an array, which will represents buildings.

more than 2 like, 3 are mandatory to ensure that water may trapped between 2 buildings



2) the elements of array should not be in either ascending or descending order.

because in this case, water will not be trapped...



so, we have to ensure these two major cases, and if they are not comes and all is good, we can move towards the next procedures...

after these two checks, the final part is comes. in this part we have to create two helping arrays called Auxiliary arrays.

one of them is used to store the left highest building for each element of array.

second is used to store the right highest building for each element of array.

our array is given : `arr[] = [4 , 2 , 0 , 6 , 3 , 2 , 5];`

each element of the array is represents the height of individual building.

let's see how to create the array containing left highest building for each element...

suppose, its name is 'leftHighestBlock'

`leftHighestBlock[0] = arr[0] // 4`

`leftHighestBlock[1] = max(leftHighestBlock[0] , arr[1]) // 4`

`leftHighestBlock[2] = max(leftHighestBlock[1] , arr[2]) // 4`

`leftHighestBlock[3] = max(leftHighestBlock[2] , arr[3]) // 4`

`leftHighestBlock[4] = max(leftHighestBlock[3] , arr[4]) // 6`

`leftHighestBlock[5] = max(leftHighestBlock[4] , arr[5]) // 6`

`leftHighestBlock[6] = max(leftHighestBlock[5] , arr[6]) // 6`

let's see how to create the array containing right highest building for each element...

suppose, its name is 'rightHighestBlock'

`rightHighestBlock[6] = arr[6] // 5`

`rightHighestBlock[5] = max(arr[5] , rightHighestBlock[6]) // 5`

`rightHighestBlock[4] = max(arr[4] , rightHighestBlock[5]) // 5`

`rightHighestBlock[3] = max(arr[3] , rightHighestBlock[4]) // 5`

`rightHighestBlock[2] = max(arr[2] , rightHighestBlock[3]) // 6`

`rightHighestBlock[1] = max(arr[1] , rightHighestBlock[2]) // 6`

`rightHighestBlock[0] = max(arr[0] , rightHighestBlock[1]) // 6`

our array is given : `arr[] = [4 , 2 , 0 , 6 , 3 , 2 , 5];`

now we have to start learning how to find the height of trapped water and at the same time the volume of the trapped water.

as we learnt before, that water can only be trapped if there is three or more buildings and the water is stored on the building between...

here, the width of each block is considered to be 1.

water height of i'th index can be found by `min(leftHighestBlock[i] , rightHighestBlock[i])`

water volume = (water height - block height) * width

negative water volume is not considered to be a valid volume, so if the volume is found to be negative, replace it with 0.

`sumOfWaterVolume = 0`

1)

`i = 0`

leftHighestBlock[i] = 4

rightHighestBlock[i] = 6

water height = min(leftHighestBlock[i] , rightHighestBlock[i]) = 4

block height = arr[i] = 4

water volume = ((4 - 4) * 1) = 0

sumOfWaterVolume = sumOfWaterVolume + water volume = 0 + 0 = 0

2)

i = 1

leftHighestBlock[i] = 4

rightHighestBlock[i] = 6

water height = min(leftHighestBlock[i] , rightHighestBlock[i]) = 4

block height = arr[i] = 2

water volume = ((4 - 2) * 1) = 2

sumOfWaterVolume = sumOfWaterVolume + water volume = 0 + 2 = 2

3)

i = 2

leftHighestBlock[i] = 4

rightHighestBlock[i] = 6

water height = min(leftHighestBlock[i] , rightHighestBlock[i]) = 4

block height = arr[i] = 0

water volume = ((4 - 0) * 1) = 4

sumOfWaterVolume = sumOfWaterVolume + water volume = 2 + 4 = 6

4)

i = 3

leftHighestBlock[i] = 4

rightHighestBlock[i] = 5

water height = min(leftHighestBlock[i] , rightHighestBlock[i]) = 4

block height = arr[i] = 6

water volume = ((4 - 6) * 1) = (-2) = 0

sumOfWaterVolume = sumOfWaterVolume + water volume = 6 + 0 = 6

5)

i = 4

leftHighestBlock[i] = 6

rightHighestBlock[i] = 5

water height = min(leftHighestBlock[i] , rightHighestBlock[i]) = 5

block height = arr[i] = 3

water volume = ((5 - 3) * 1) = 2

$\text{sumOfWaterVolume} = \text{sumOfWaterVolume} + \text{water volume} = 6 + 2 = 8$

6)

$i = 5$

$\text{leftHighestBlock}[i] = 6$

$\text{rightHighestBlock}[i] = 5$

$\text{water height} = \min(\text{leftHighestBlock}[i], \text{rightHighestBlock}[i]) = 5$

$\text{block height} = \text{arr}[i] = 2$

$\text{water volume} = ((5 - 2) * 1) = 3$

$\text{sumOfWaterVolume} = \text{sumOfWaterVolume} + \text{water volume} = 8 + 3 = 11$

7)

$i = 6$

$\text{leftHighestBlock}[i] = 6$

$\text{rightHighestBlock}[i] = 5$

$\text{water height} = \min(\text{leftHighestBlock}[i], \text{rightHighestBlock}[i]) = 5$

$\text{block height} = \text{arr}[i] = 5$

$\text{water volume} = ((5 - 5) * 1) = 0$

$\text{sumOfWaterVolume} = \text{sumOfWaterVolume} + \text{water volume} = 11 + 0 = 11$